

# Rodríguez-Huerta, Edgar

Rights Lab Senior Research Fellow in Social Sustainability and Complex Systems

Data Science, Sustainability and Supply Chain Mapping | £2.1m in Co-I & PI grant funding | FWCI: 1.42

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driguez/ · Nottingham, UK

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## Experience

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| Oct 2024 – present  | <b>Rights Lab Senior Research Fellow in Social Sustainability and Complex Systems</b> University of Nottingham, Rights Lab – Nottingham, UK <ul style="list-style-type: none"><li>Quantitative research focused on understanding climate change and decent work interconnections.</li><li>Supply chain mapping and analysis to generate risk assessments using Social-LCA.</li><li>Generation of climate change models to explore effect on labour conditions.</li></ul> |
| Dec 2021 – Sep 2024 | <b>Rights Lab Research Fellow and Lead in Social-Ecological Systems Modelling</b> University of Nottingham, Rights Lab – Nottingham, UK <ul style="list-style-type: none"><li>Awarded Global Talent Visa by UK Government with British Academy endorsement</li></ul>                                                                                                                                                                                                     |
| Dec 2021 – Sep 2024 | <b>Rights Lab Research Fellow in Sustainable Ecosystems and Society</b> University of Nottingham, Rights Lab – Nottingham, UK                                                                                                                                                                                                                                                                                                                                            |

## Industry Experience

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| Jan 2010 – Jan 2013 | <b>Senior Project Developer and Business Intelligence Specialist</b> JLC Corporation – Mérida, México <ul style="list-style-type: none"><li>Managed Business Intelligence and Enterprise Resource Program software, including stakeholder training</li><li>Designed dashboards and reports for retail, financial and academic sectors.</li></ul> |
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## Education

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| Sep 2015 – Mar 2020 | <b>Ph.D. in Sustainability Science</b> Universitat Politècnica de Catalunya, Institute for Sustainability Science – Barcelona, Spain <ul style="list-style-type: none"><li>Dissertation: Water Societal Metabolism in Yucatan Peninsula, Mexico. Application of Multi-Scale Integrated Analysis of Societal and Ecosystem Metabolism (MuSI-ASEM)</li><li>Three peer-reviewed publications as first author</li></ul> |
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- Built exploratory tool to analyse effect of climate change in water demand (R, QGIS, Tableau)

**Master's degree in Sustainability Science and Technology** Universitat Politècnica de Catalunya, Institute for Sustainability Science — Barcelona, Spain

- 120 ECTS. Grade: 8.9/10. Rank 2/12
- MSc thesis: Societal Metabolism in the Energy Sector in Catalunya, applying spatial analysis with QGIS
- Collaborated with multi-cultural and interdisciplinary team

**Master's degree in Logistics** Universidad de Zaragoza, Zaragoza Logistics Center — Zaragoza, Spain

- 105 ECTS. Grade: 9.1/10
- MSc thesis: Methodologies for a Sustainability Balanced Scorecard in the Logistics Sector
- Professional fellowship program in logistics. Training program in multiple companies

**Undergraduate degree in Industrial Engineering** Instituto Tecnológico de Mérida — Mérida, México

## Selected Publications

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Peer reviewed

- Rodríguez-Huerta, E., Leão L., Landman, T. (2025). Climate change, decent work and workers' health in Brazil: theoretical considerations. *Revista Brasileira de Saúde Ocupacional*. <https://doi.org/10.1590/2317-6369/16224en2025v50eddsst12>
- Rodríguez-Huerta, E., Bell, B., Jackson, B., Blackstone, N.T., Battaglia, K., Marquez, A.S., Benoît Norris, C., Decker Sparks, J.L., Conrad, Z., Matteson, J. (2025). Current and recommended diets in the USA have embedded forced labour risk. *Nature Food*. <https://www.nature.com/articles/s43016-025-01242-8>
- Tigchelaar, M., Jackson, B., Selig, E., Davis, A., O'Regan, E., Trond, K., Nakayama, S., Boyd, D., Cheung, W., Rodríguez-Huerta, E., Williams, C., Decker Sparks, J. (2025). Conceptualization of Decent Work in Fishing in a Changing Climate. *Marine Policy*. <https://doi.org/10.1016/j.marpol.2025.106846>
- Boyd, D., Jackson, B., Decker Sparks, J., Giles M. F., Girindran, R., Gosling, S., Trodd, Z. Ni Bhriain, L., Rodríguez-Huerta, E. (2024). The future of decent work: Forecasting heat stress and the intersection of sustainable development challenges in India's brick kilns. *Sustainable Development*. <https://doi.org/10.1002/sd.3272>
- Blackstone, N.T., Battaglia K., Rodríguez-Huerta, E., Bell, B., Decker Sparks J., Cash S., Conrad Z., Nikkhah A., Jackson B., Matteson J., Gao S., Fuller K., Zhang F.F., Webb P. (2024). Diets cannot be sustainable without ensuring the well-being of communities, workers, and animals in food value chains. *Nature Food*. <https://doi.org/10.1038/s43016-024-01048-0>
- Lumley-Sapanski A., Rodríguez-Huerta, E., Young, M., Nicholson A., Schwarz K. (2024). Criminalizing survivors of modern slavery: the United Kingdom's National Referral Mechanism as a border-making process. *Journal of Social Policy*. <https://doi.org/10.1017/S0047279424000230>
- Blackstone, N.T., Rodríguez-Huerta, E., Battaglia K., Jackson B., Jackson E., Benoît Norris C., Decker Sparks, J.L. (2023). Forced labour risk is pervasive in the US land-based food supply. *Nature Food*. <https://doi.org/10.1038/s43016-023-00794-x>

- Rodríguez-Huerta, E., Rosas-Casals, M., Hernandez-Terronez, L.M (2020). A water balance model to estimate climate change impact on groundwater recharge in Yucatan Peninsula, Mexico. *Hydrological Science Journal*. <https://www.tandfonline.com/doi/full/10.1080/02626667.2019.1702989>
- Rodríguez-Huerta, E., Rosas-Casals, M., Hernandez-Terronez, L.M (2019). Water societal metabolism in the Yucatan Peninsula. The impact of climate change on the recharge of groundwater by 2030. *Journal of Cleaner Production*. <https://www.sciencedirect.com/science/article/pii/S0959652619322875>
- Rodríguez-Huerta, E., Rosas-Casals, M., Sorman A.H. (2017). A societal metabolism approach to job creation and renewable energy transitions in Catalonia. *Energy Policy*. <https://www.sciencedirect.com/science/article/pii/S0301421517303786>

#### Reports

- Rodríguez-Huerta, E., Domingos Martinez dos Santos, I., de Almeida Moura, F., Faria Leal, C.R., Landman, T., Castellero, I.T.A., Coutinho, K.G., Moraes, L., Pavia, L.R., Manuella Gallego, Soares, M.R., Brandão, M.P.F., Barros Costa, S., Siviero, A.A., Trevizan, A.F., Galvão Gomes, P.I. (2026). Climate Change, Decent Work, and Workers' Health: Vulnerabilities and Responses of Workers in Brazilian Agriculture. *CLIDEWO project*. <https://doi.org/10.5281/zenodo.18839271>
- Landman, T., Brotherton V., Bosatta L., Rodríguez-Huerta, E. (2025). Towards a Just Transition in the UK food retail sector. *Sainsbury's and Rights Lab human rights research partnership*. <https://www.nottingham.ac.uk/research/beacons-of-excellence/rights-lab/resources/reports-and-briefings/2025/september/briefing-paper-just-transition-in-the-uk-food-retail-sector.pdf>
- Cockayne, J., Rodríguez-Huerta E., Burcu, O. (2022). The Energy of Freedom? Solar Energy, Modern Slavery and the Just Transition. *The British Academy*. <https://www.thebritishacademy.ac.uk/publications/the-energy-of-freedom-solar-energy-modern-slavery-and-the-just-transition/>
- Boyd, D., Rodríguez-Huerta, E., Jackson B., Decker Sparks, J.L. (2021). The Social and Ecological Impacts of Supply Chains. *Research report*.

## Grants and Fellowships

|             |                                                                                                                    |
|-------------|--------------------------------------------------------------------------------------------------------------------|
| 2025 – 2026 | <b>UK Home Office (Modern Slavery Innovation Fund)</b>                                                             |
|             | The GeoAI for Decent Work: Innovating Engagement with Government, Civil Society and Survivors                      |
| 2024 – 2025 | <b>British Academy (ISPF ODA Challenge-Oriented Research Grants)</b>                                               |
|             | Climate Change, Occupational Health, and Decent Work: Worker Vulnerabilities and Response in Brazilian Agriculture |
| 2024 – 2025 | <b>UK Home Office (Modern Slavery Innovation Fund)</b>                                                             |
|             | Adding value to the GeoAI Platform for Decent Work: A Survivor-Informed Socio-Ecological Assessment                |
| 2024 – 2025 | <b>Industry collaboration (Retail sector)</b>                                                                      |
|             | Salient and Evolving Human Rights Risks                                                                            |
| 2024 – 2025 | <b>UK Home Office (Modern Slavery Innovation Fund)</b>                                                             |
|             | Using Data Science to Transform India's Brick Kilns                                                                |
| 2021 – 2022 | <b>British Academy (Just Transitions within Sectors and Industries Globally)</b>                                   |
|             | Solar Energy, Modern Slavery, and the Just Transition                                                              |

2021 – 2022

**World Wild Fund (USA)**

The Social and Ecological Impacts of Supply Chains

2014 – 2020

**Consejo Nacional de Ciencia y Tecnología (CONACyT - México)**

PhD and MSc Scholarship

2014 – 2015

**Agència de Gestió d'Ajuts Universitaris i de Recerca (AGAUR)**

MSc Fellowship for Directed Academic Activities

## Awards

2025

**Distinguished alumni of 2025 in the Industrial Engineering programme**

Instituto Tecnológico de Mérida

2008

**Chamber of Commerce of Aragon, Spain — Memoria Pilot award** Report in Logistics Excellence for large enterprises (La Bella Easo, Zaragoza)

2026

**Queen Elizabeth Prize for Education** Team member of Slavery from Space: Using Satellites for Human Rights and Sustainable Development

2022

**Global Talent Visa for Exceptional Promise** UK Government with the British Academy endorsement

## Selected Events and Collaborations

Jul 2025

**II International Congress on Ecological Humanities** The Johns Hopkins University - Universitat Pompeu Fabra Public Policy Center — Barcelona, Spain

Decent Work and Workers Health nexus in Brazilian agriculture

Jun 2025

**Just transitions in food, farming and fisheries** Ethical Trading Initiative, IIED, BananaLink — London, UK

Apr 2025

**Addressing Human Trafficking and Forced Labour in the Anthropocene** Survivor Alliance, The Human Trafficking Legal Center, The Freedom Fund — Oxford, UK

Sep 2024

**Visualizing Complexity Science** Complexity Science Hub — Vienna, Austria

2021 – 2024

**Friedman School of Nutrition Science and Policy at Tufts University** Leading A Sustainability Transition in Nutrition Globally (LASTING) Project — Boston, MA, USA

Leading A Sustainability Transition in Nutrition Globally (LASTING) Project — Boston, MA, USA

May 2023

**Decent Fisheries Work in a Changing Climate** Stanford University, the King Center and Center for Ocean Solutions — Stanford, CA, USA

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| Mar 2022 | <b>How Social Life Cycle Assessment (S-LCA) can help address forced labour</b><br>U.S. Department of Labor and Tufts University — Washington DC, USA                                                                                                           |
| Feb 2021 | How Social Life Cycle Assessment (S-LCA) can help address forced labour<br><b>22nd Ordinary Session of the Yucatan Peninsula Basin Council</b> Consejo de Cuenca de la Península de Yucatán & Ithaca Environmental — Mérida, México                            |
| Feb 2026 | Invitation to co-design the Regional Water Program 2020-2024 of the Yucatan Peninsula<br><b>Decent Work and Workers Health nexus in Brazilian agriculture</b> National Workshop in Brasília Debates Climate Change, Health, and Labor in Brazilian Agriculture |
| Jun 2026 | Climate Risk Vulnerability Assessment for Agriculture workers<br><b>Sustainability science to tackle forced labour: methods, measurements and case studies</b> Universitat Politècnica de Catalunya, Terrassa, Catalunya                                       |
| Jul 2026 | <b>I International Seminar on Human Trafficking: Empirical Evidence, Measurement, and New Theoretical Debates</b> organized by the Federal University of Pernambuco, Brazil                                                                                    |

## Teaching

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| Nov 2021 | <b>Data management and visualization for academic research</b> Instituto Tecnológico de Mérida — Mérida, México                                                                        |
|          | Graduate course                                                                                                                                                                        |
|          | • Course for MSc students in Regional Development Studies                                                                                                                              |
| Jul 2025 | <b>Seminar: Sustainability science to tackle forced labour: methods and measurements</b> Universitat Politècnica de Catalunya, Institute for Sustainability Science — Barcelona, Spain |
|          | Seminar                                                                                                                                                                                |

## Selected Extra Training

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| Jun 2025 | <b>Research Communication Programme</b> Research Academy University of Nottingham                                                                                      |
| Apr 2025 | <b>MOOC: Partnering for change — Link research to societal challenges</b><br>Network for Transdisciplinary Research td-net of the Swiss Academies of Arts and Sciences |
| Feb 2024 | <b>Stepping into Leadership 2023/24 — LMA Development Programmes</b> Research Academy University of Nottingham                                                         |
| Oct 2023 | <b>MOOC: Analysing/Presenting Data/Information</b> Edward Tufte Course (8h)                                                                                            |

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| Jan 2021 | <b>MOOC: Data Science — Statistics and Machine Learning Specialization</b><br>Coursera Programme (Johns Hopkins University) (133h)                              |
| Oct 2020 | <b>MOOC: Specialization in Data Science (R Programming)</b> Coursera Programme<br>(Johns Hopkins University) (290h)                                             |
| Sep 2020 | <b>MOOC: Spatial Data Science and Applications</b> Coursera Programme (Yonsei<br>University) (24h)                                                              |
| Aug 2020 | <b>MOOC: Systems Thinking in Public Health</b> Coursera Programme (Johns Hop-<br>kins University) (20h)                                                         |
| Jul 2020 | <b>MOOC: Ecosystem Services — A Method for Sustainable Development</b><br>Coursera Programme (University of Geneva) (24h)                                       |
| Jul 2020 | <b>MOOC: Sustainability of Social-Ecological Systems — Nexus among Water,<br/>Energy, and Food</b> Coursera Programme (Universitat Autònoma de Barcelona) (48h) |

## Skills

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| Programming<br>Software | R, SQL (intermediate), Quarto<br>Tableau [Desktop and Prep], Rstudio, QGIS, OpenLCA, Zotero, VosViewer, Xerte<br>(intermediate), Affinity (learning), Observable (learning) |
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## Languages

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| Spanish | Native |
| English | Fluent |

## Academic Service

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| Grants and Journals Re-<br>viewed | British Academy for Knowledge Frontiers: Just Transition; Journal of Cleaner<br>Production; Energy Policy; Hydrogeology Journal; Joule; Anti-Trafficking Review;<br>Economía Creativa.                                                                                                                                                                                                                            |
| Student Supervision               | PhD Students: 3 supervised — Topics in Food Systems. MSc Students: 4 supervised<br>— Research in data extraction, and regional development. Support for MSc students<br>from the Technical University of Denmark (DTU) for the project “Sociotechnical<br>dimension of renewable energies” (Apr 2024). BSc Students: 3 supervised — Place-<br>ment students related to data science projects and just transition. |
| Societies                         | Member of The British Academy Early Career Researcher Network since 2023.                                                                                                                                                                                                                                                                                                                                         |